

☒ HARD

☐ ADVANCED MATH

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# Advanced Math

# 09

10 questions · ~15 min

Q1

ADVANCED MATH

The expression  $\frac{x^{-2}y^{\frac{1}{2}}}{x^{\frac{1}{3}}y^{-1}}$ , where  $x > 1$  and  $y > 1$ , is equivalent to which of the following?

- A.  $\frac{\sqrt{y}}{\sqrt[3]{x^2}}$
- B.  $\frac{y\sqrt{y}}{\sqrt[3]{x^2}}$
- C.  $\frac{y\sqrt{y}}{x\sqrt{x}}$
- D.  $\frac{y\sqrt{y}}{x^2\sqrt[3]{x}}$

$$y = x^2 - 14x + 22$$

The given equation relates the variables  $x$  and  $y$ . For what value of  $x$  does the value of  $y$  reach its minimum?

STUDENT-PRODUCED RESPONSE

|   |   |   |   |
|---|---|---|---|
|   |   |   |   |
| . | / | . | . |
| 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 |

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$64x^2 + bx + 25 = 0$$

In the given equation,  $b$  is a constant. For which of the following values of  $b$  will the equation have more than one real solution?

- A. -91
- B. -80
- C. 5
- D. 40

$$(ax + 3)(5x^2 - bx + 4) = 20x^3 - 9x^2 - 2x + 12$$

The equation above is true for all  $x$ , where  $a$  and  $b$  are constants. What is the value of  $ab$  ?

- A. 18
- B. 20
- C. 24
- D. 40

The functions  $f$  and  $g$  are defined by the given equations, where  $x \geq 0$ . Which of the following equations displays, as a constant or coefficient, the maximum value of the function it defines, where  $x \geq 0$ ?

I.  $fx = 330.4^{x+3}$

II.  $gx = 330.160.4^{x-2}$

- A. I only
- B. II only
- C. I and II
- D. Neither I nor II

$$\frac{1}{x^2 + 10x + 25} = 4$$

If  $x$  is a solution to the given equation, which of the following is a possible value of  $x+5$  ?

- A.  $\frac{1}{2}$
- B.  $\frac{5}{2}$
- C.  $\frac{9}{2}$
- D.  $\frac{11}{2}$

$$\sqrt[3]{p^2} = t^{\frac{9}{7}}$$

In the given equation,  $p > 1$  and  $t > 1$ . If  $t = p^{3n-1}$ , where  $n$  is a constant, what is the value of  $n$ ?

STUDENT-PRODUCED RESPONSE

|   |   |   |   |
|---|---|---|---|
|   |   |   |   |
| . | / | . | . |
| 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 |

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



$$fx = x + 7^2 + 4$$

The function  $f$  is defined by the given equation. For what value of  $x$  does  $f(x)$  reach its minimum?

STUDENT-PRODUCED RESPONSE

| . | / . | / . | . |
|---|-----|-----|---|
| 0 | 0   | 0   | 0 |
| 1 | 1   | 1   | 1 |
| 2 | 2   | 2   | 2 |
| 3 | 3   | 3   | 3 |
| 4 | 4   | 4   | 4 |
| 5 | 5   | 5   | 5 |
| 6 | 6   | 6   | 6 |
| 7 | 7   | 7   | 7 |
| 8 | 8   | 8   | 8 |
| 9 | 9   | 9   | 9 |

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$\frac{x^2}{\sqrt{x^2 - c^2}} = \frac{c^2}{\sqrt{x^2 - c^2}} + 39$$

In the given equation,  $c$  is a positive constant. Which of the following is one of the solutions to the given equation?

- A.  $-c$
- B.  $-c^2 - 39^2$
- C.  $-\sqrt{39^2 - c^2}$
- D.  $-\sqrt{c^2 + 39^2}$

The function  $f$  is defined by  $f(x) = a\sqrt{x+b}$ , where  $a$  and  $b$  are constants. In the  $xy$ -plane, the graph of  $y = f(x)$  passes through the point  $(-24, 0)$ , and  $f(24) < 0$ . Which of the following must be true?

- A.  $f(0) = 24$
- B.  $f(0) = -24$
- C.  $a > b$
- D.  $a < b$